Software Engineering (SE) – Practical / Oral Examination

Question & Answer Set | Academic Year 2026–27

Vidyavardhini's College of Engineering & Technology, Department of Computer Engineering

Topics covered in this document

Topic	Questions
UML diagrams – Use Case, Class, Statechart, Activity, Component	Q1, Q2, Q3, Q4, Q12, Q20
Effort / cost estimation – Basic COCOMO, Function Points, LOC method	Q5, Q18, Q19
Project scheduling – Gantt chart	Q6
Data Flow Diagrams – Level 0 and Level 1	Q7, Q16, Q17
Software process models – Agile, Waterfall, Spiral, Incremental	Q8, Q13, Q14, Q15
White-box / Basis Path Testing – Cyclomatic Complexity	Q9, Q10, Q11

Q 1

Draw the Use Case Diagram for a Library Management System.

Answer:

Actors: Librarian, Member (student / staff), Admin. **System boundary:** Library Management System.

- **Member:** search, reserve, renew books and pay fine.
- **Librarian:** issue and return books, manage books and members.
- **Admin:** manage books and members, generate reports.

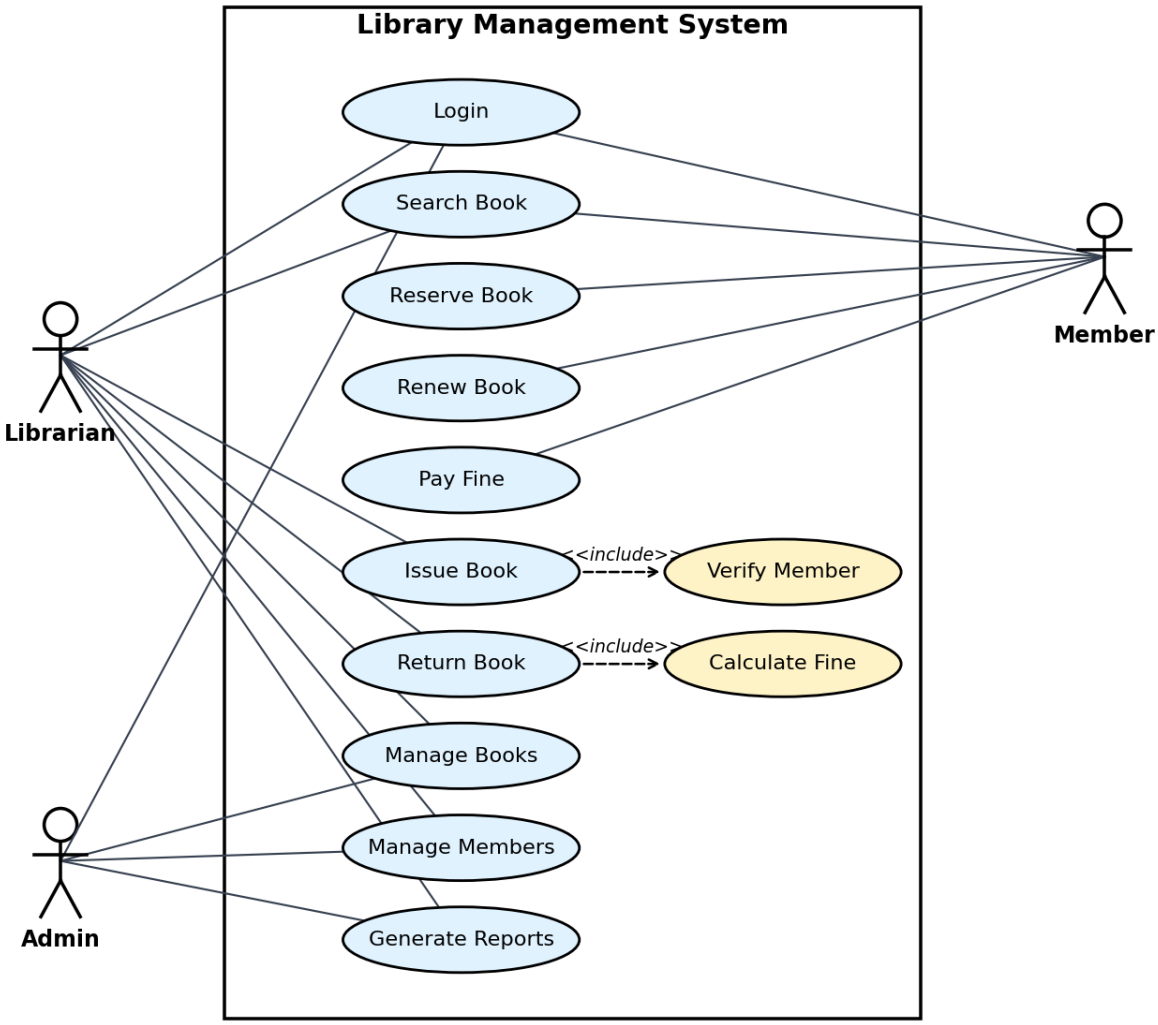


Fig. Q1 – Use Case Diagram: Library Management System

Q 2

Draw the Class Diagram for a Hospital Management System.

Answer:

Patient and Doctor inherit from Person. A Department has many Doctors, a Patient has one MedicalRecord, and Patients have Appointments, Bills and Lab Reports.

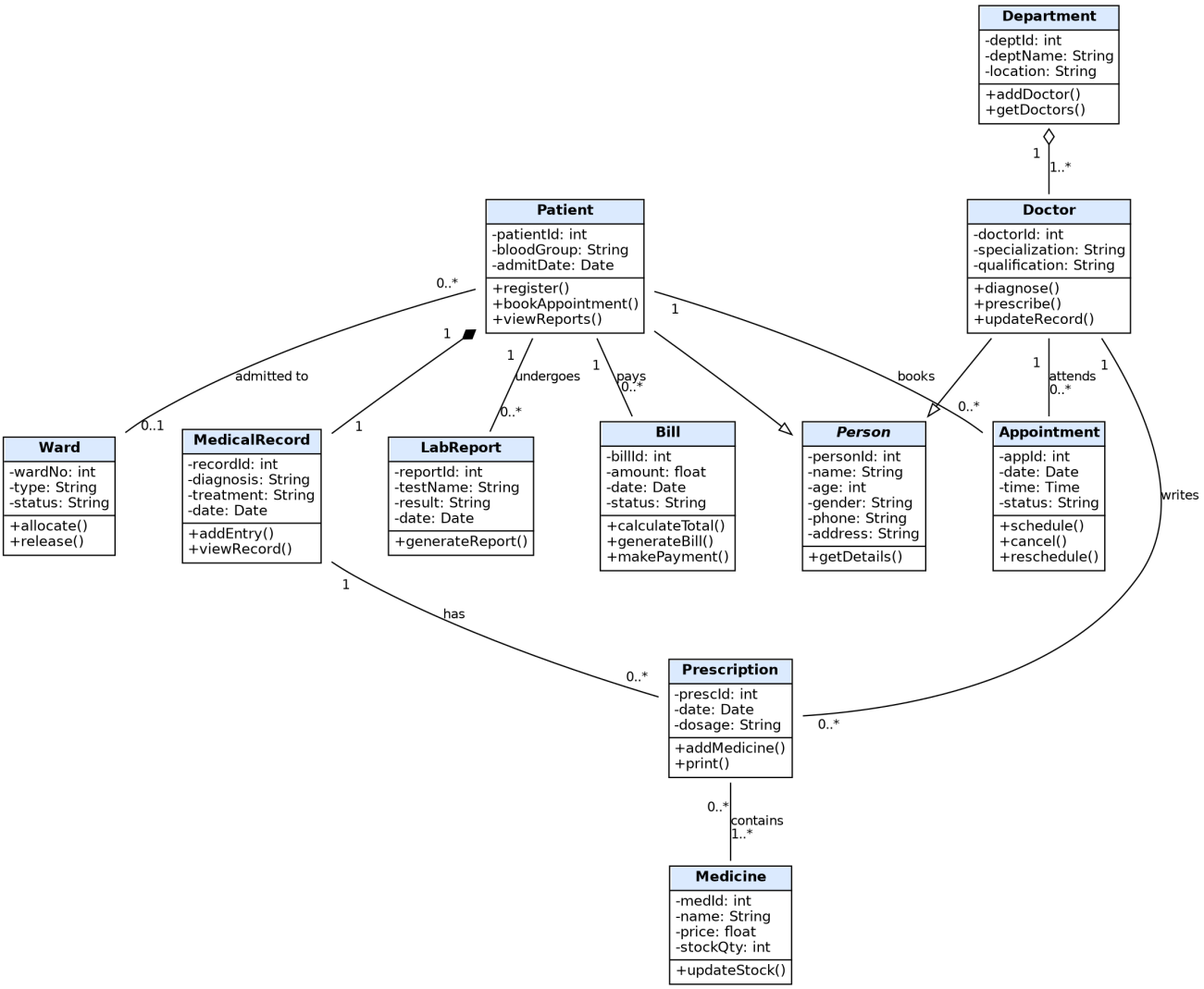


Fig. Q2 – Class Diagram: Hospital Management System

Q 3

Draw the Statechart Diagram for a Online shopping cart

Answer:

The cart starts Empty, becomes Active when items are added, then goes through Checkout and Payment. The order ends as Delivered or Cancelled.

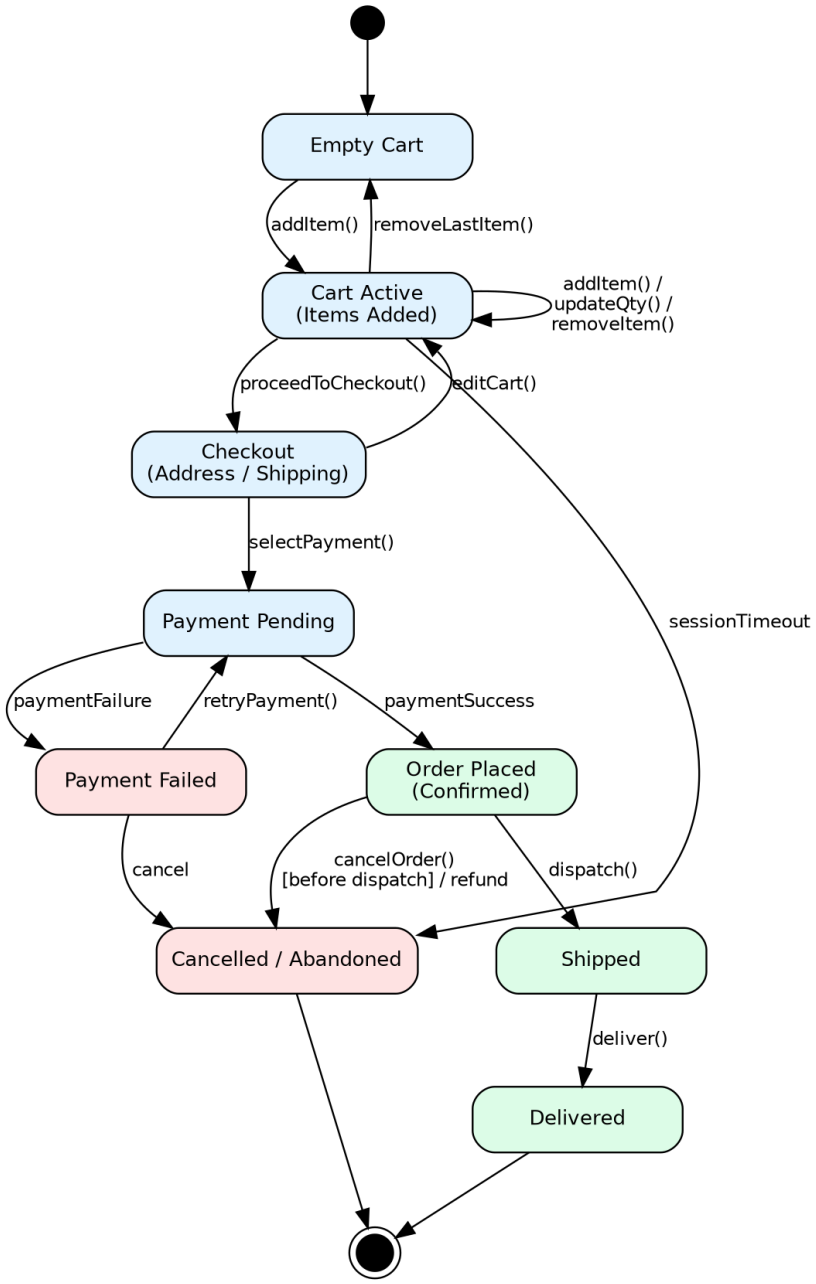


Fig. Q3 – Statechart Diagram: Online Shopping Cart

Q 4

Draw the Activity Diagram for a student Management System.

Answer:

Login, choose an operation (details, attendance, results, fees, reports), validate data, update the database, show confirmation, then repeat or logout.

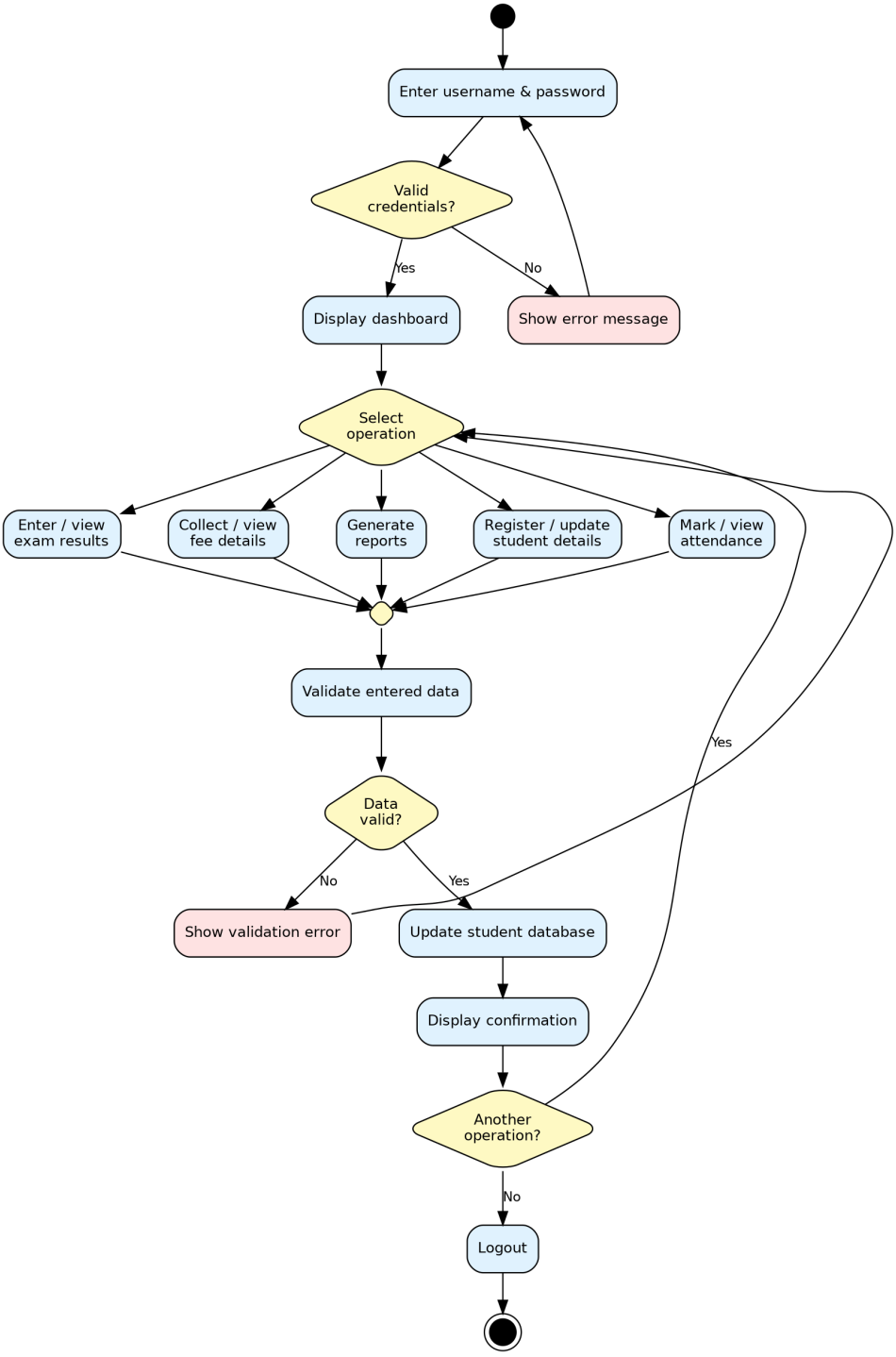


Fig. Q4 – Activity Diagram: Student Management System

Q 5

Estimate the development effort and time for a project of size 20 KLOC using the Basic COCOMO model. Assume the project is Organic, with (a=2.4, b=1.05, c=2.5, d=0.38). Calculate the estimated project cost if the development cost is ₹30,000 per person-month.

Answer:

Formulas: Effort $E = a \times (KLOC)^b$ person-months; Development time $D = c \times (E)^d$ months.

Step 1 – Effort: $E = 2.4 \times (20)^{1.05} = 2.4 \times 23.23 =$ **55.76 person-months**

Step 2 – Time: $D = 2.5 \times (55.76)^{0.38} = 2.5 \times 4.61 =$ **11.52 months**

Step 3 – Average staff: $N = E / D = 55.76 / 11.52 =$ **4.84 ≈ 5 persons**

Step 4 – Cost: Cost = E × cost per person-month = 55.76 × 30,000 = **₹16,72,684** (≈ ₹16.73 lakh)

Parameter	Value
Effort	55.76 person-months
Development time	11.52 months
Average team size	4.84 persons
Estimated cost	₹16,72,684

Q 6

A college plans to develop a Student Management System to manage student registration, attendance, examination results, and fee details.

1. Identify the major project activities and assign suitable durations.
2. Define task dependencies and prepare a project schedule using a Gantt chart

Answer:

1. Major activities and durations / 2. Dependencies

ID	Activity	Duration	Depends on	Start	Finish
A	Requirement gathering & SRS	3 wk	–	Week 1	Week 3
B	System design (architecture, UI, DB)	3 wk	A	Week 4	Week 6
C	Database setup	2 wk	B	Week 7	Week 8
D	Registration module	3 wk	C	Week 9	Week 11
E	Attendance module	3 wk	C	Week 9	Week 11
F	Examination results module	3 wk	C	Week 9	Week 11
G	Fee management module	3 wk	C	Week 9	Week 11
H	Integration of modules	2 wk	D,E,F,G	Week 12	Week 13
I	System & user acceptance testing	3 wk	H	Week 14	Week 16
J	Deployment, training & handover	1 wk	I	Week 17	Week 17

Modules D–G are independent after the database is ready, so they run **in parallel** (weeks 9–11). **Critical path:** A → B → C → (D/E/F/G) → H → I → J. **Total duration = 17 weeks.**

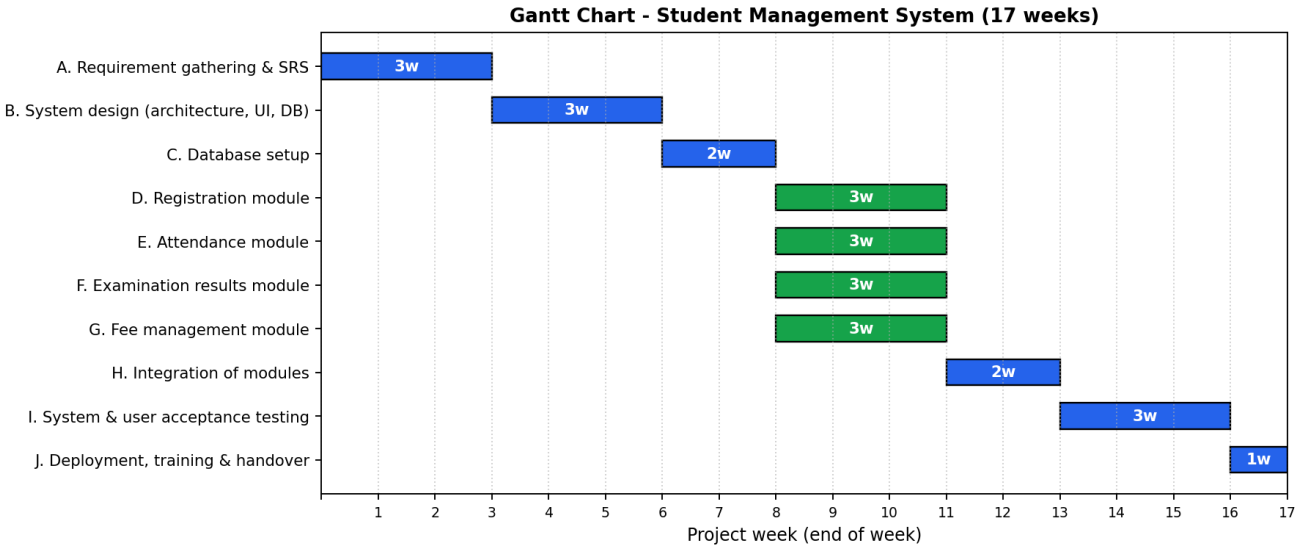


Fig. Q6 – Gantt Chart: Student Management System (blue = sequential tasks, green = parallel modules)

Q 7

Consider a Hostel Management System that manages student hostel admissions, room allocation, fee payments, and complaints.

Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

Answer:

External entities: Student, Warden, Accounts Office. **Processes:** 1.0 Admission Processing, 2.0 Room Allocation, 3.0 Fee Management, 4.0 Complaint Handling, 5.0 Report Generation. **Data stores:** D1 Student Records, D2 Room Details, D3 Fee Records, D4 Complaint Records.

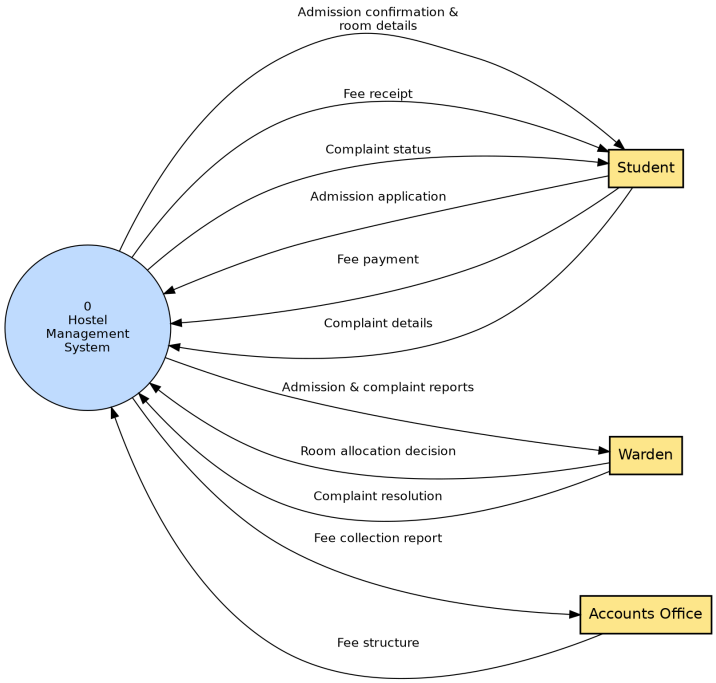


Fig. Q7(a) – Level 0 (Context) DFD: Hostel Management System

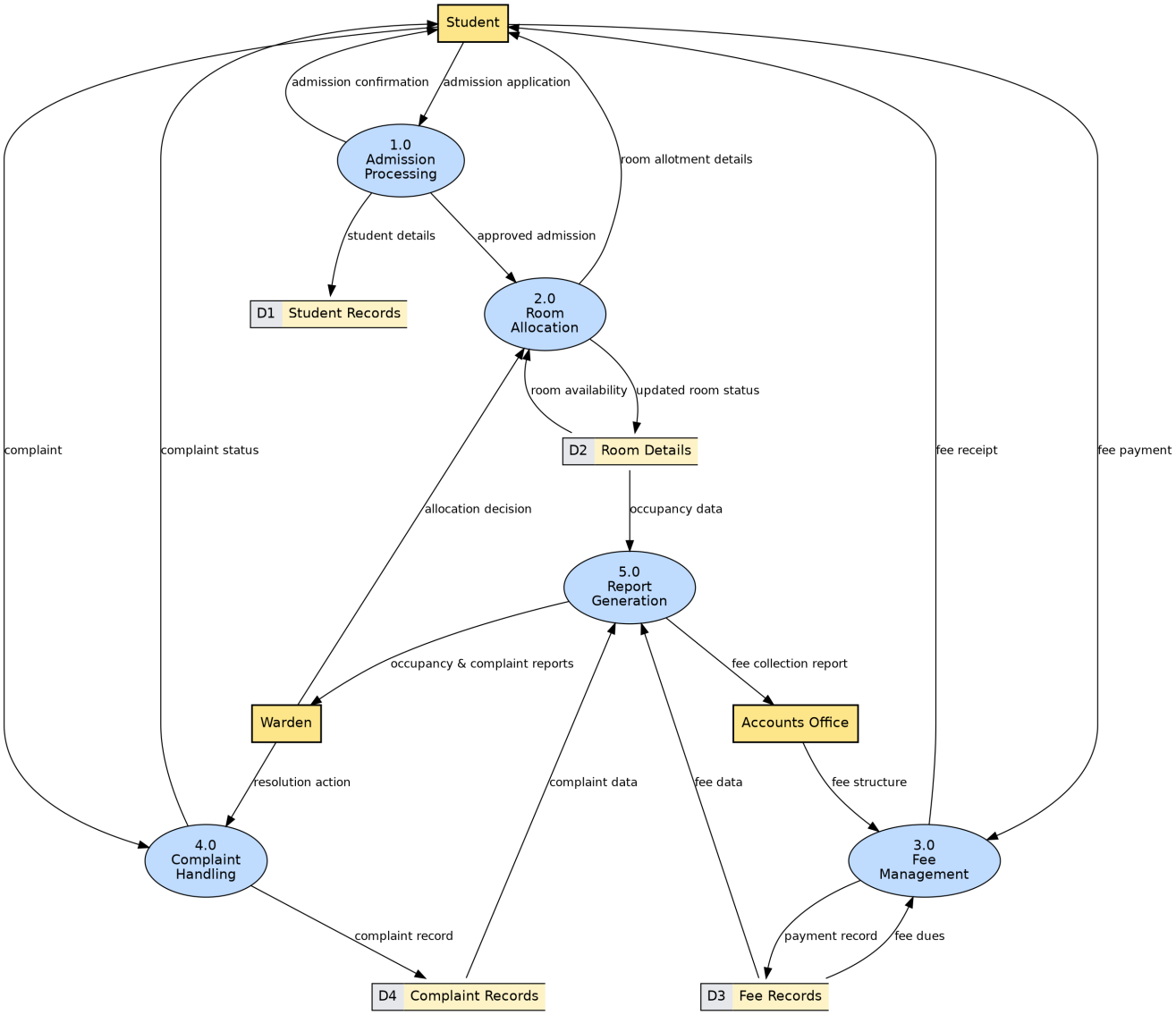


Fig. Q7(b) – Level 1 DFD: Hostel Management System

Q 8

A college plans to develop an Online Examination System to conduct online tests, evaluate answers, generate results, and maintain student records. The requirements may change based on feedback from faculty and students. Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

Answer:

Suitable model: Agile (Scrum) – iterative and incremental development.

- Requirements change with feedback – Agile handles change in every sprint.
- Modules are delivered as small working increments.
- Waterfall is unsuitable because it freezes requirements early.

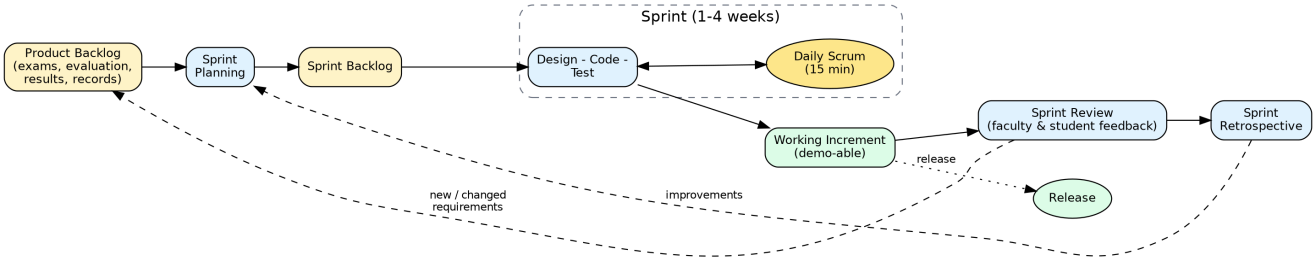


Fig. Q8 – Agile (Scrum) Process Model

Q 9

1. Apply Basis Path Testing to the given program and calculate the Cyclomatic Complexity.

read n; if (n > 0) { if (n % 2 == 0) print "Even" else print "Odd" } else print "Invalid"

Answer:

Node	Statement
1	read n
2	if (n > 0)
3	if (n % 2 == 0)
4	print "Even"
5	print "Odd"
6	print "Invalid"
7	Exit (join point)

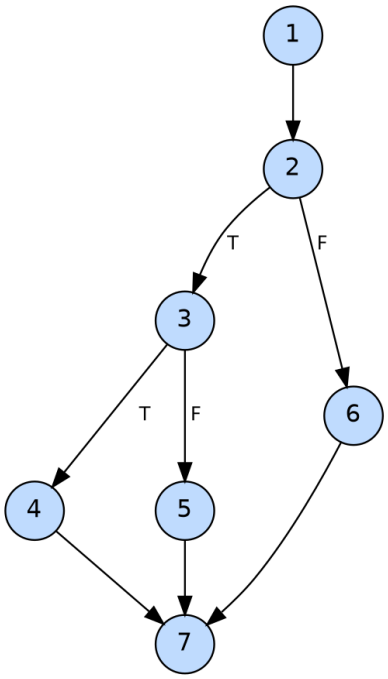


Fig. Q9 – Control Flow Graph

Cyclomatic Complexity

Nodes N = 7, Edges E = 8, Predicate nodes P = 2 (nodes 2 and 3), Regions R = 3.

V(G) = E − N + 2 = 8 − 7 + 2 = **3** (also P + 1 = 3)

Independent paths and test cases

Path	Nodes	Test input	Expected output
P1	1-2-3-4-7	n = 4	Even
P2	1-2-3-5-7	n = 7	Odd
P3	1-2-6-7	n = −3 (or 0)	Invalid

Q 10

Apply White-box testing and calculate cyclomatic complexity for given program

read n; i = 2; isPrime = true; while (i <= n/2) { if (n % i == 0) { isPrime = false; break } i = i + 1 } print isPrime

Answer:

Node	Statement
1	read n; i = 2; isPrime = true
2	while (i <= n/2)
3	if (n % i == 0)
4	isPrime = false; break
5	i = i + 1
6	print isPrime

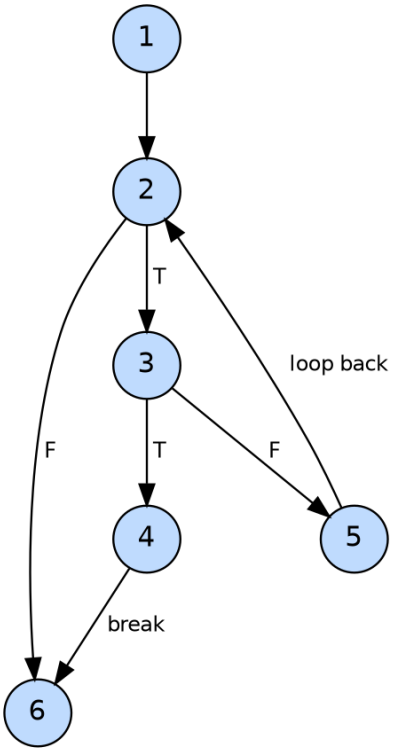


Fig. Q10 – Control Flow Graph

Cyclomatic Complexity

N = 6, E = 7, P = 2 (while at node 2, if at node 3), R = 3.

V(G) = E − N + 2 = 7 − 6 + 2 = **3** (also P + 1 = 3)

White-box test cases (basis paths)

Path	Nodes	Test input	Expected output
P1 (loop skipped)	1-2-6	n = 3	true
P2 (divisor found, break)	1-2-3-4-6	n = 4	false

Path	Nodes	Test input	Expected output
P3 (one iteration, no divisor, exit)	1-2-3-5-2-6	n = 5	true

Since V(G) = 3, a minimum of 3 test cases are needed to cover every independent path.

Q 11

Given the following pseudo-code, apply Basis Path Testing to determine:

read a, b; if (a > b) max = a else max = b; if (max > 0) print "Positive" else print "Non-positive"

1. Control Flow Graph (CFG) 2. Cyclomatic Complexity 3. Independent Paths 4. Test Cases

Answer:

1. Control Flow Graph

Node	Statement
1	read a, b
2	if (a > b)
3	max = a
4	max = b
5	if (max > 0)
6	print "Positive"
7	print "Non-positive"
8	Exit

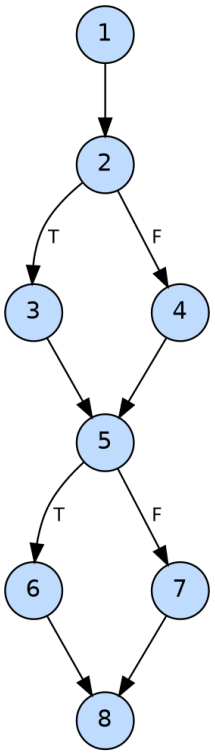


Fig. Q11 – Control Flow Graph

2. Cyclomatic Complexity

N = 8, E = 9, P = 2 (nodes 2 and 5), R = 3.

V(G) = E − N + 2 = 9 − 8 + 2 = **3** (also P + 1 = 3)

3. Independent paths (basis set)

- **Path 1:** 1-2-3-5-6-8 (a > b, max > 0)
- **Path 2:** 1-2-4-5-6-8 (a ≤ b, max > 0)
- **Path 3:** 1-2-3-5-7-8 (a > b, max ≤ 0)

(The fourth possible path 1-2-4-5-7-8 is a combination of the above, hence not independent.)

4. Test cases

Path	Input (a, b)	Expected output
1	(5, 3)	Positive
2	(3, 5)	Positive
3	(−1, −5)	Non-positive

Q 12

Draw the Component Diagram for a Library Management System.

Answer:

Three layers: UI, business components (login, search, members, issue/return, fines, notifications, reports) and the database. Dashed arrows show dependencies.

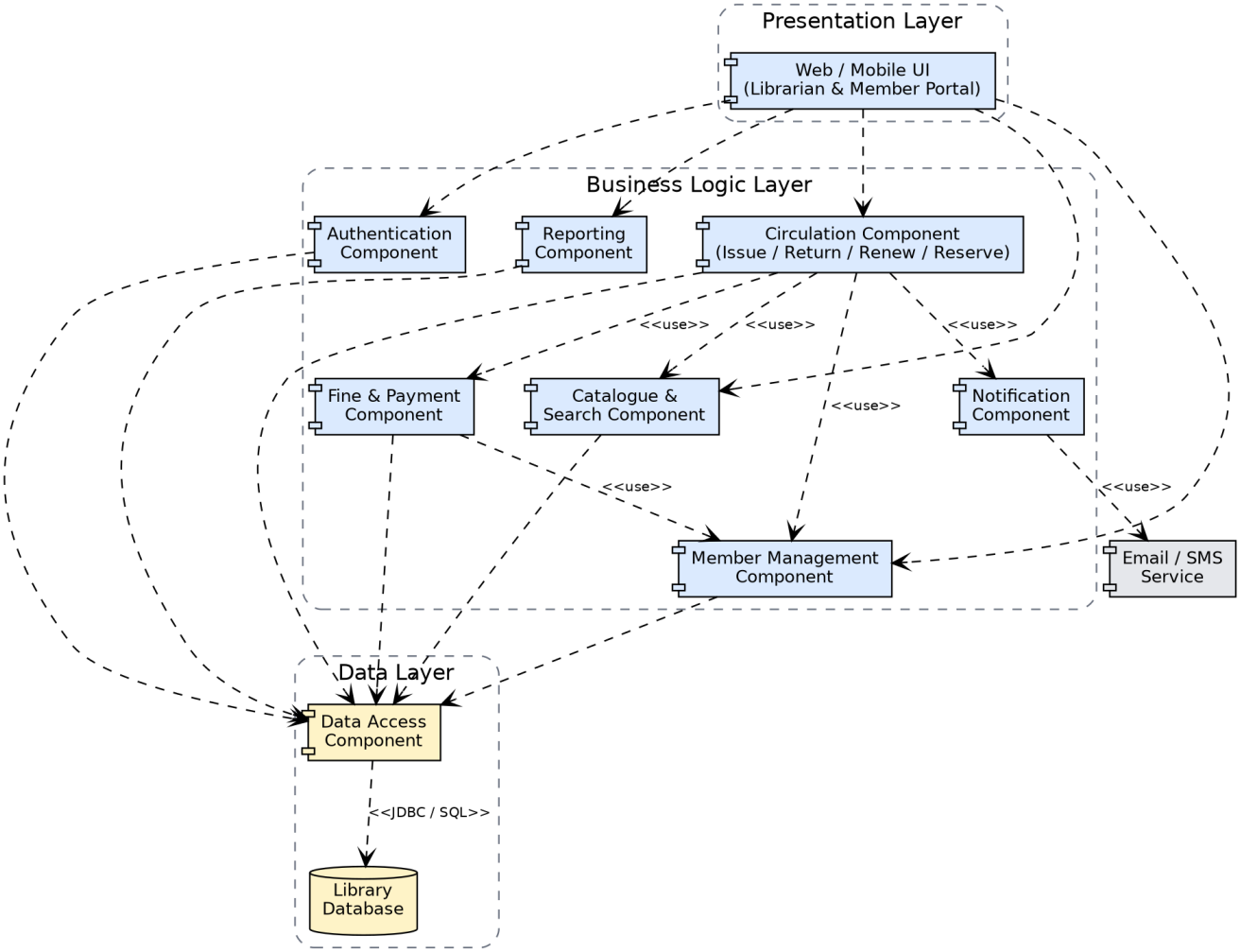


Fig. Q12 – Component Diagram: Library Management System

Q 13

Hospital Management System – A hospital plans to develop a Hospital Management System to manage patient registration, appointments, billing, laboratory reports, pharmacy, and electronic health records. The hospital requires a secure and reliable system with clearly defined requirements. Major requirement changes are not expected during development.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

Answer:

Suitable model: Waterfall (Classical Life-Cycle) Model.

- Requirements are clear and stable.
- Needs strong documentation and review after every phase.
- Changes are not expected, so Waterfall's rigidity is acceptable.

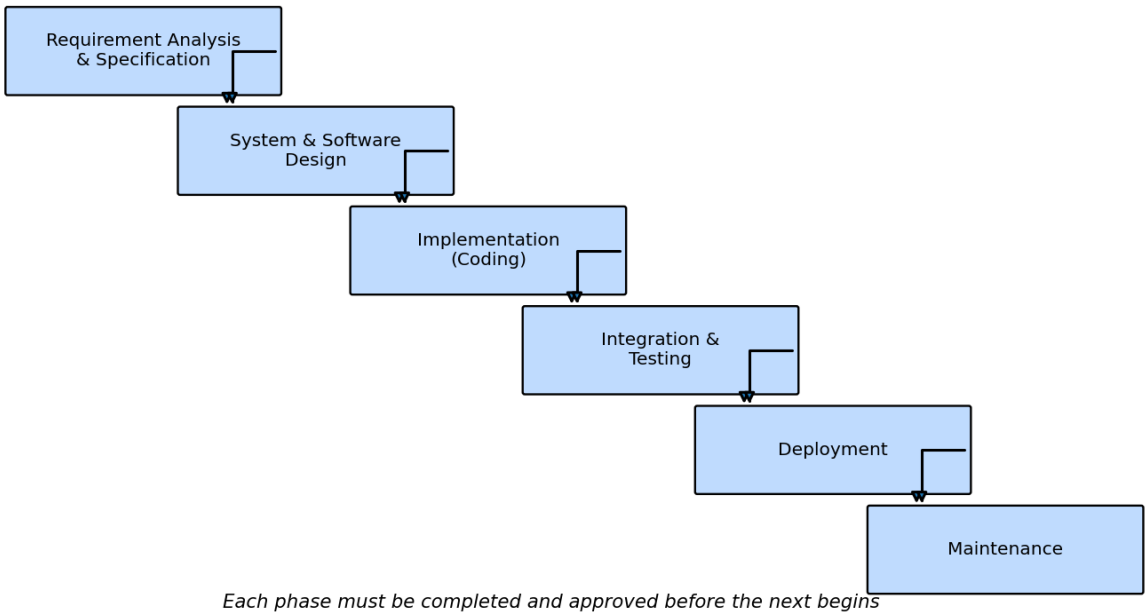


Fig. Q13 – Waterfall Model

Q 14

Mobile Banking Application – A bank plans to develop a Mobile Banking Application with fund transfer, balance enquiry, bill payment, investment services, and biometric authentication. Since financial transactions are involved, the system requires high security, extensive testing, and risk management.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

Answer:

Suitable model: Spiral Model.

- Every loop has a risk analysis step – important for security and fraud.
- Prototypes are tested repeatedly before release.
- Built in stages with a go / no-go review each loop.

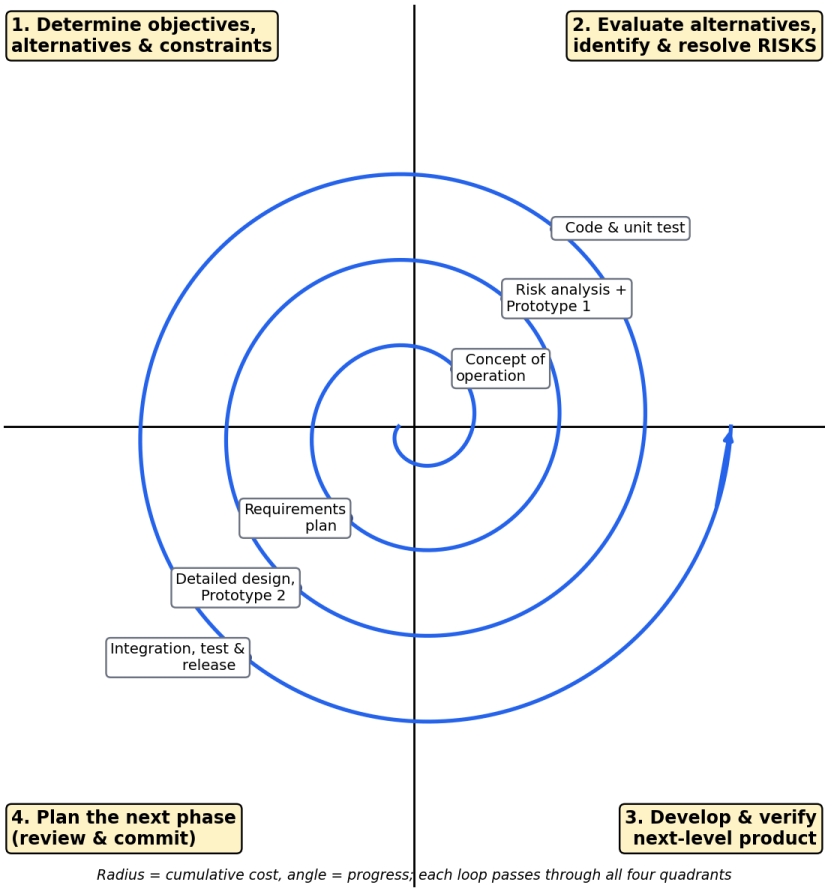


Fig. Q14 – Spiral Model

Q 15

AI-Based Smart Traffic Management System – A city corporation plans to develop an AI-based Smart Traffic Management System to monitor traffic, optimize signal timings, detect accidents, and provide real-time route suggestions. The project involves AI integration, IoT devices, and evolving requirements.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

Answer:

Suitable model: Iterative–Incremental Model (with prototyping, in an Agile style).

- Requirements evolve, so it is delivered in increments: monitoring → signal optimisation → accident detection.
- Early prototypes validate the AI and IoT parts.
- Feedback from each release improves the next one.

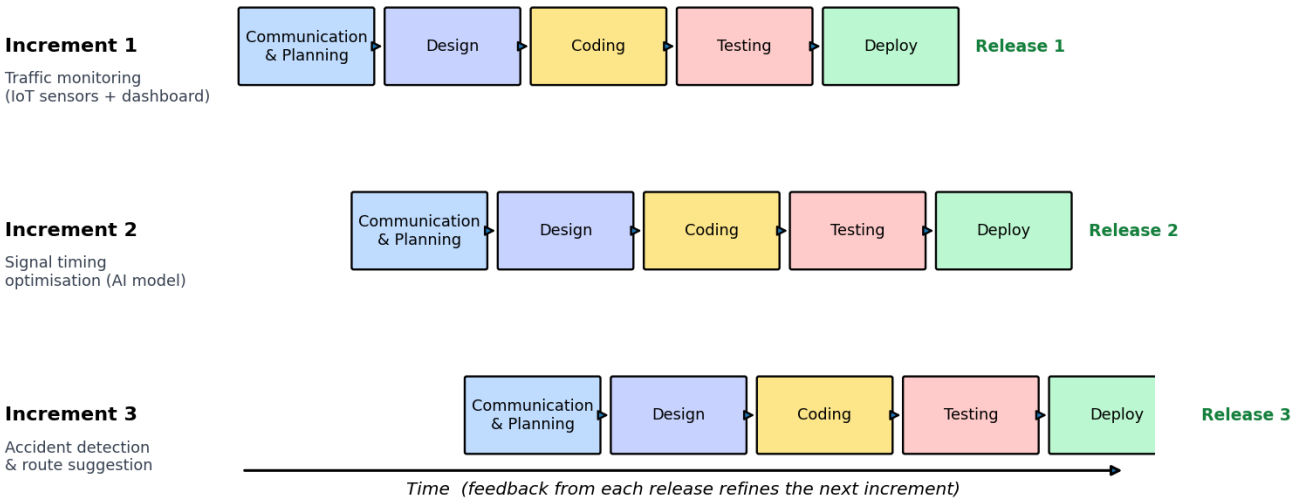


Fig. Q15 – Iterative–Incremental Model

Q 16

Consider course registration system

Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

Answer:

External entities: Student, Faculty, Administrator. **Processes:** 1.0 Authenticate & Manage Profile, 2.0 Manage Course Catalogue, 3.0 Register for Courses, 4.0 Verify Fee Payment, 5.0 Generate Reports & Timetable. **Data stores:** D1 Student Records, D2 Course Records, D3 Registration Records, D4 Fee Records.

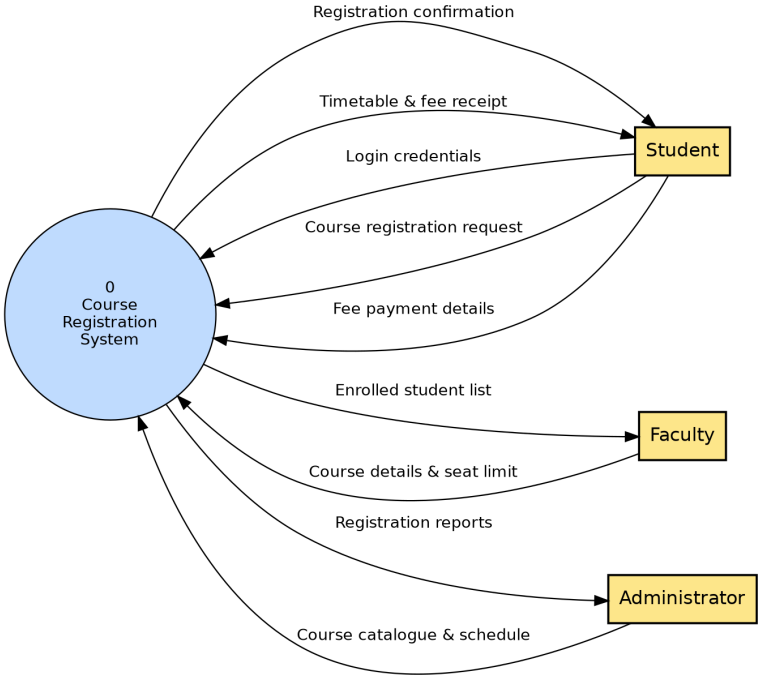


Fig. Q16(a) – Level 0 (Context) DFD: Course Registration System

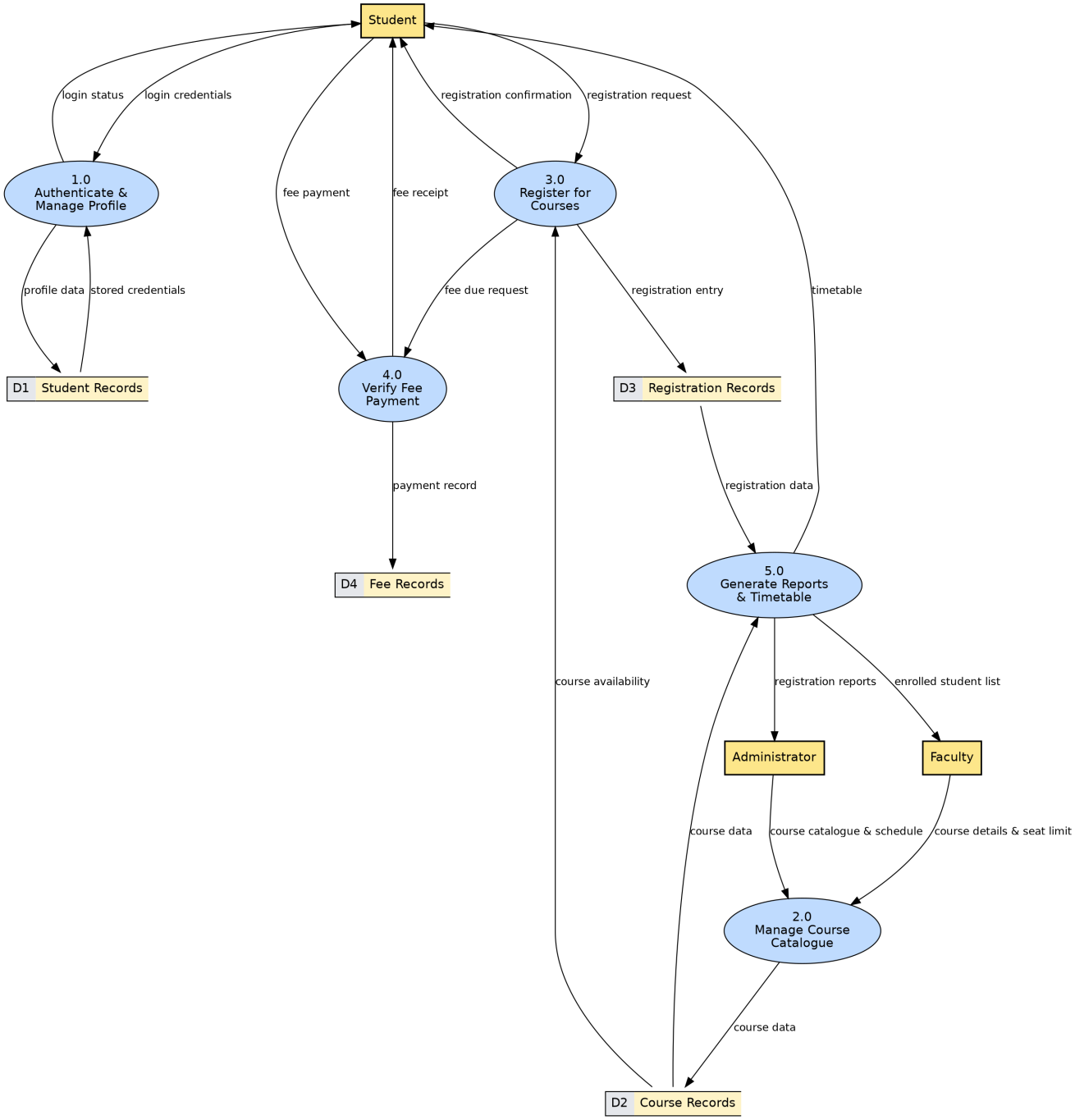


Fig. Q16(b) – Level 1 DFD: Course Registration System

Q 17

Consider railway reservation system

Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

Answer:

External entities: Passenger, Railway Admin, Bank / Payment Gateway. **Processes:** 1.0 Register/Login, 2.0 Search Trains & Check Availability, 3.0 Book Ticket, 4.0 Process Payment, 5.0 Cancel Ticket & Refund, 6.0 Manage Trains & Reports. **Data stores:** D1 User Records, D2 Train Details, D3 Booking Records, D4 Payment Records.

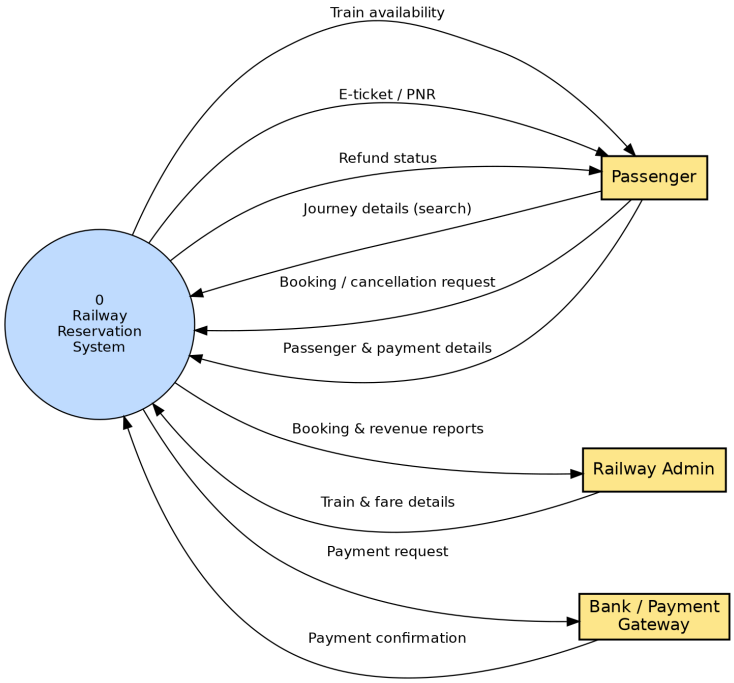


Fig. Q17(a) – Level 0 (Context) DFD: Railway Reservation System

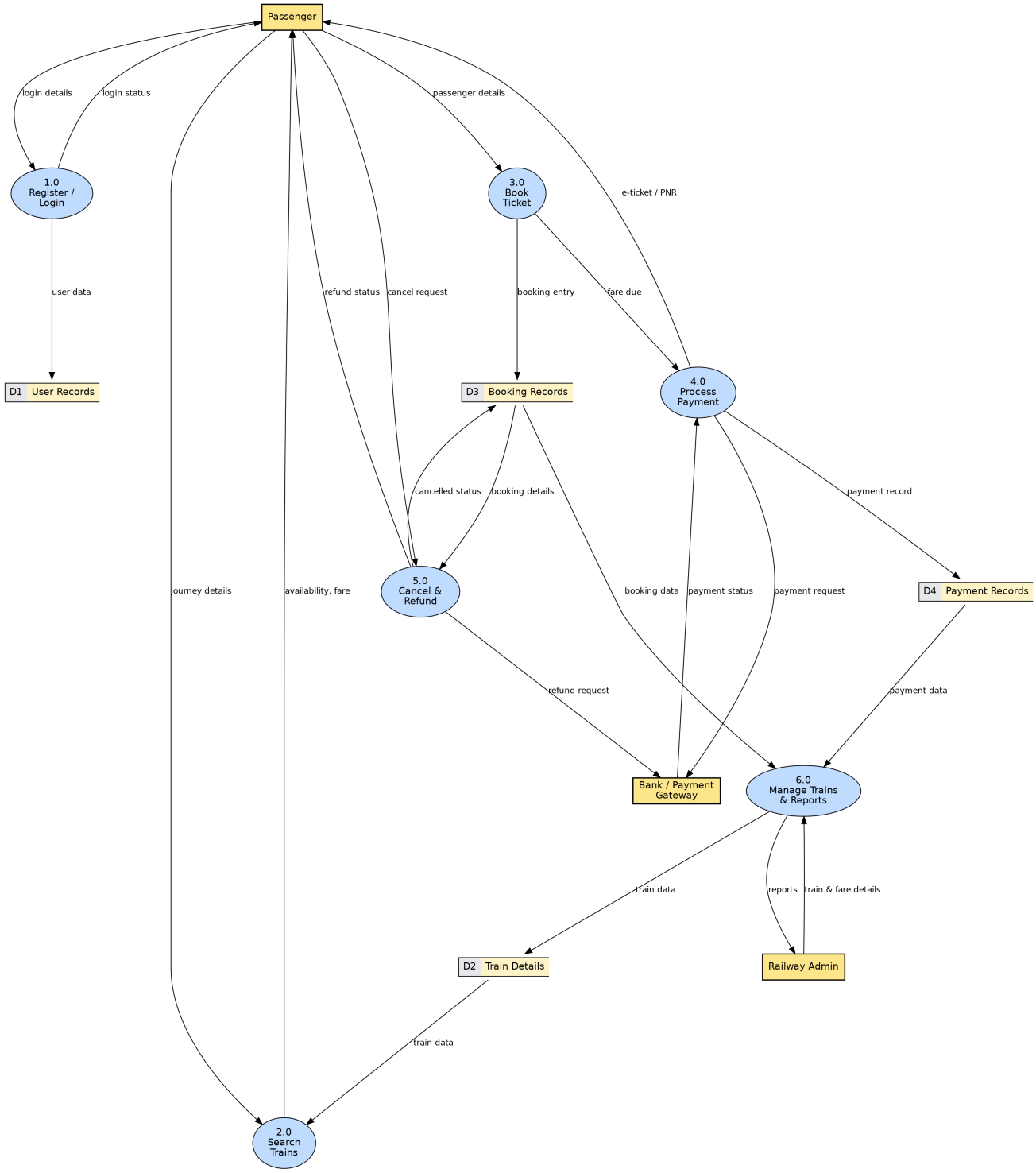


Fig. Q17(b) – Level 1 DFD: Railway Reservation System

Q 18

A healthcare startup plans to develop an Online Pharmacy Management System (search medicines, upload prescriptions, orders, online payments, delivery tracking; pharmacists verify prescriptions, manage inventory, generate sales reports). Estimated size = 32 KLOC, Semi-Detached project. EI = 18, EO = 14, EQ = 10, ILF = 8, EIF = 5; all average complexity; VAF = 1.05.

1. Calculate UFP. 2. Calculate AFP. 3. Calculate the project effort using Basic COCOMO (a=2.4, b=1.05, c=2.5, d=0.38). 4. Find the estimated effort and discuss whether the project is suitable for the selected COCOMO category.

Answer:

1. Unadjusted Function Points (average weights: EI 4, EO 5, EQ 4, ILF 10, EIF 7)

Component	Count	Weight	Count × Weight
External Inputs (EI)	18	4	72
External Outputs (EO)	14	5	70
External Inquiries (EQ)	10	4	40
Internal Logical Files (ILF)	8	10	80
External Interface Files (EIF)	5	7	35
UFP			**297**

2. Adjusted Function Points

AFP = UFP × VAF = 297 × 1.05 = **311.85 FP**

3. Basic COCOMO effort (given constants, size = 32 KLOC)

$E = 2.4 \times (32)^{1.05} = 2.4 \times 38.05 =$ **91.33 person-months**

$D = 2.5 \times (91.33)^{0.38} =$ **13.90 months**; average staff = E / D = **6.57 ≈ 7 persons**

4. Estimated effort and suitability of the category

Estimated effort ≈ 91.3 person-months (about 13.9 months with ~7 people), using the constants supplied in the question.

- The given constants (2.4, 1.05, 2.5, 0.38) are actually the *Organic* constants. Standard Semi-Detached constants (3.0, 1.12, 2.5, 0.35) give 145.5 person-months and 14.3 months.
- Semi-Detached suits a medium project with mixed team experience and moderately complex requirements. This system (prescriptions, payments, delivery, inventory) fits that, so the category is suitable – although 32 KLOC is a small size.
- So the realistic estimate is the higher Semi-Detached value; the Organic result is a lower bound.

Q 19

A municipal corporation plans to develop a Smart Parking Management System (locate parking spaces, reserve slots, online payments, real-time updates; admins monitor occupancy and generate daily reports). Size = 28 KLOC, Organic. a=2.4, b=1.05, c=2.5, d=0.38.

1. Calculate the project effort and development time using Basic COCOMO.

2. If the organization's productivity is 600 LOC per person-month, calculate the effort using the LOC method.

Answer:

1. Basic COCOMO

Effort $E = 2.4 \times (28)^{1.05} = 2.4 \times 33.08 =$ **79.38 person-months**

Development time $D = 2.5 \times (79.38)^{0.38} =$ **13.18 months**

Average staff = E / D = 79.38 / 13.18 ≈ **6.0 ≈ 6 persons**

2. LOC method

Effort = Size / Productivity = 28,000 LOC ÷ 600 LOC per person-month = **46.67 person-months**

COCOMO (79.4 PM) is higher than the LOC method (46.7 PM) because COCOMO assumes larger projects are less productive per person.

Q 20

Draw Class diagram for Travel Agency system

Answer:

Classes: Customer, Agent, TourPackage, Destination, Booking, Flight, Hotel, Payment.

- Customer 1 – 0..* Booking (makes); Agent 1 – 0..* Booking (handles) and Agent 1 – 0..* TourPackage (creates).
- Booking 0..* – 1 TourPackage; Booking 1 – 1..* Payment (composition).
- TourPackage covers 1..* Destinations, and includes 0..* Flights and 0..* Hotels (aggregation).

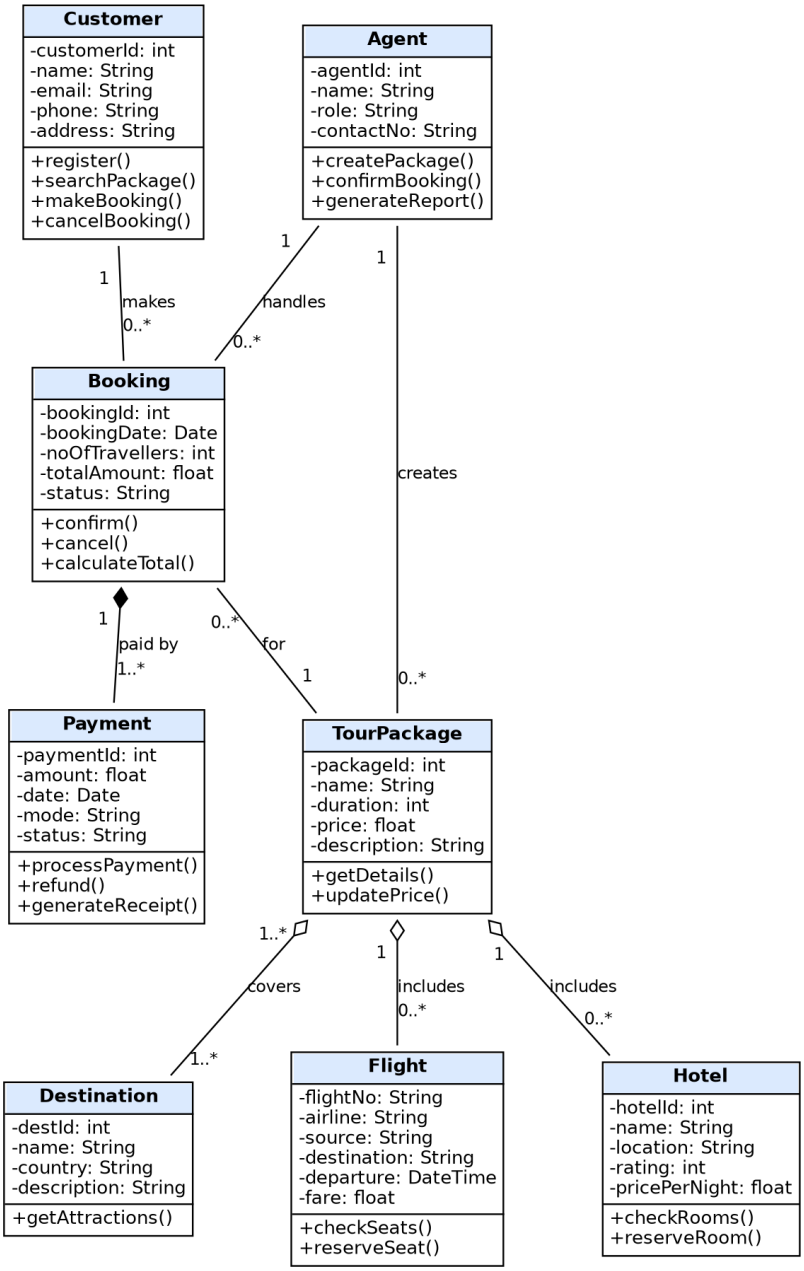


Fig. Q20 – Class Diagram: Travel Agency System